

How to Determine the Logical Mapping

1. Understanding Logical Mapping

Objective

We focus on a specialized class of SVGs known as **Logical Charts**, which faithfully encode the analytical richness of data using SVG primitives. The central aim is to establish a precise **Logical Coordinate System**, defined by the data range $[d_{\min}, d_{\max}]$, and to determine its exact correspondence to the **User Coordinate System**, defined by the viewBox range $[vb_{\min}, vb_{\max}]$.

For two dimensions, the viewBox is typically expressed as:

$$[x_{vb_{\min}}, y_{vb_{\min}}, x_{vb_{\max}} - x_{vb_{\min}}, y_{vb_{\max}} - y_{vb_{\min}}]$$

We also propose a **Local Logical Mapping** for transformed coordinate systems, and introduce a new extension to SVG—**SVGX**—which formalizes logical mappings.

Principle of Mapping

The mapping relies on **linear proportionality** between the Logical and User coordinate systems. This guarantees mathematical exactness in both directions.

Required Values from the SVG Document

To calculate the logical mapping, extract the following:

1. Two Reference Points

Identify two axis labels with known value-to-coordinate correspondences:

- **Point 1:** Logical value d_1 and user-space coordinate v_1
- **Point 2:** Logical value d_2 and user-space coordinate v_2

2. ViewBox Boundaries

Extract the full viewBox dimensions:

$$[vb_{\min}, vb_{\max}]$$

In two dimensions:

$$[x_{vb_{\min}}, y_{vb_{\min}}, x_{vb_{\max}} - x_{vb_{\min}}, y_{vb_{\max}} - y_{vb_{\min}}]$$

2. Deriving the Mapping

Using proportionality, we calculate a logical value d_x based on its known user-space coordinate v_x :

General Mapping Formula

$$\frac{d_x - d_1}{d_2 - d_1} = \frac{v_x - v_1}{v_2 - v_1}$$

Solving for d_x :

$$d_x = d_1 + (v_x - v_1) \cdot \frac{d_2 - d_1}{v_2 - v_1}$$

Compute Logical Extremes

Logical Minimum ($d_{\min}$)

Calculated from the **maximum** visual coordinate ($vb_{\max}$):

$$d_{\min} = d_1 + (vb_{\max} - v_1) \cdot \frac{d_2 - d_1}{v_2 - v_1}$$

Logical Maximum ($d_{\max}$)

Calculated from the **minimum** visual coordinate ($vb_{\min}$):

$$d_{\max} = d_1 + (vb_{\min} - v_1) \cdot \frac{d_2 - d_1}{v_2 - v_1}$$

3. Global Logical Mapping

Once derived, we obtain a complete logical mapping:

$$[d_{\min}, d_{\max}, vb_{\min}, vb_{\max}]$$

4. Local Logical Mapping

Whenever there is a SVG transformation with `<g>` tag, we have a local coordinate system in action. We have a local logical mapping in logical chart SVG. In such cases, apply the same procedure as above to establish a local logical mapping:

$$[d_{\min}, d_{\max}, vg_{\min}, vg_{\max}]$$

5. SVGX: A Semantic Extension for Logical Charts

We formally propose **SVGX**, an extension to the SVG specification to encode logical mappings directly.

Syntax Specification

Global Mapping

```
<svg
  xlm = [d_{x_min}, d_{x_max}, x_{vb_min}, x_{vb_max}],
  ylm = [d_{y_min}, d_{y_max}, y_{vb_min}, y_{vb_max}]
>
```

Local Mapping

```
<g
  xlm = [d_{x_min}, d_{x_max}, x_{vg_min}, x_{vg_max}],
  ylm = [d_{y_min}, d_{y_max}, y_{vg_min}, y_{vg_max}]
>
```

6. Why SVGX Matters

SVGX introduces formal semantic control that's missing from conventional SVG workflows. By embedding logical domain mappings directly into the markup, it:

- Enables exact domain-to-visual alignment
- Supports modular, nested visual components
- Facilitates runtime transformation and responsive layout
- Establishes declarative animation semantics tied to logical values

SVGX can serve as the semantic backbone for scalable visual intelligence systems, making SVG not just render-rich but **meaning-aware**.